④ (Inductive step)

$(v_1, \ldots, v_{m-1}, v_m)$ : linearly independent

$\Leftarrow$ want

① $(u_1, \ldots, u_{m-1}, u_m)$ : orthonormal

② $R = \begin{array}{c c} & \overbrace{\phantom{R'}}^{m-1} \\ m-1\{ & \begin{bmatrix} R' & * \\ 0 & a_{m,m} \end{bmatrix} \end{array} \in M_{m \times m}(\mathbb{R})$

s.t. $[v_1 \ldots v_m] = [u_1, \ldots, u_m] R$.

Consider the simplest case where

$(v_1, \ldots, v_{m-1}) = (u_1, \ldots, u_{m-1})$

so that $R' = I_{m-1}$

want:

$[u_1 \ldots u_{m-1}, \underline{v_m}] = [u_1, \ldots, u_m] \begin{bmatrix} I_{m-1} & * \\ 0 & a_{m,m} \end{bmatrix}$

want:

$$v_m = a_{1,m}u_1 + a_{2,m}u_2 + \dots + a_{m-1,m}u_{m-1} + \underbrace{a_{m,m}}_{\neq 0}\boxed{u_m}$$

$$\Leftrightarrow \quad u_m = b_{1,m}u_1 + b_{2,m}u_2 + \dots + b_{m-1,m}u_{m-1} + b_{m,m}v_m$$

s.t. (1) $u_m \cdot u_i = 0 \quad \forall i < m$

(2) $u_m \cdot u_m = 1$

For (1): $u_m \cdot u_i \quad \forall i < m$

$$= (b_{1,m}u_1 + \dots + b_{m-1,m}u_{m-1} + b_{m,m}v_m) \cdot u_1$$

$$= b_{1,m}(u_1 \cdot u_1) + b_{2,m}(u_2 \cdot u_1) + \dots + b_{m-1,m}(u_{m-1} \cdot u_1) + b_{m,m}(v_m \cdot u_1)$$

$$= b_{i,m} + b_{m,m}(v_m \cdot u_i) = 0$$

$$\Rightarrow \boxed{r_{i,m} = -r_{m,m}(v_m \cdot u_i)}$$

For ②:

$$u_m \cdot u_m = 1$$

$$\parallel$$

$$\left[r_{1,m} u_1 + \dots + r_{m-1,m} u_{m-1} + r_{m,m} v_m\right]$$

$$\cdot \left[r_{1,m} u_1 + \dots + r_{m-1,m} u_{m-1} + r_{m,m} v_m\right]$$

$$= r_{1,m}^2 + r_{2,m}^2 + \dots + r_{m-1,m}^2$$

$$+ 2 r_{m,m} \left[r_{1,m}(u_1 \cdot v_m) + r_{2,m}(u_2 \cdot v_m) + \dots + r_{m-1,m}(u_{m-1} \cdot v_m)\right]$$

$$+ r_{m,m}^2 (v_m \cdot v_m)$$

positive multiple of $r_{m,m}^2$

Choose $r_{m,m}$ so that this is 1.

# Gram-Schmidt & QR decomposition

Example: $n=3$

$(V_1, V_2, V_3)$ : linearly independent

Step 1 : $u_1 = \dfrac{V_1}{\|V_1\|}$

Want

Step 2: $u_2 = r_{1,2} u_1 + r_{2,2} V_2$

s.t.
- $u_1 \cdot u_2 = 0$
- $u_2 \cdot u_2 = 1$

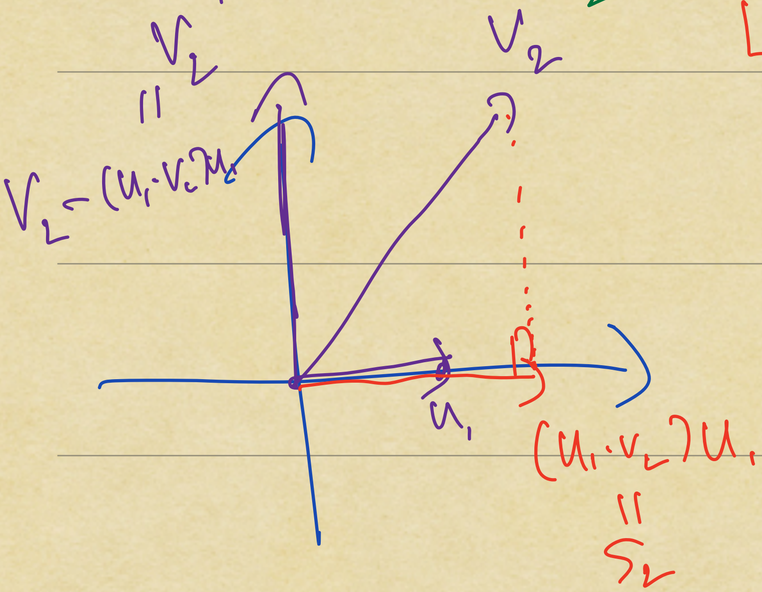
$$0 = u_1 \cdot u_2 = u_1 \cdot (r_{1,2} u_1 + r_{2,2} V_2)$$
$$= r_{1,2} + r_{2,2} (u_1 \cdot V_2)$$

$$\Rightarrow r_{1,2} = -r_{2,2} (u_1 \cdot V_2)$$

Initial guess: $r_{2,2} = 1 \Rightarrow r_{1,2} = -(u_1 \cdot v_2)$

$$v_2' = -(u_1 \cdot v_2)u_1 + v_2$$

$$= v_2 - (u_1 \cdot v_2)u_1$$




$v_2 = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ $\quad (v_2 \cdot u_1) = a_1$

$u_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\quad \Downarrow$

$(v_2 \cdot u_1)u_1 = \begin{bmatrix} a_1 \\ 0 \end{bmatrix}$

$v_2'$ might not have length 1

Take $u_2 = \dfrac{v_2'}{\|v_2'\|}$

Step 3: $\underset{\text{Solve:}}{u_3} = r_{1,3}u_1 + r_{2,3}u_2 + r_{3,3}v_3$

s.t. (1) $u_3 \cdot u_1 = 0 \Leftrightarrow r_{1,3} = -r_{3,3}(v_3 \cdot u_1)$

② $u_3 \cdot u_2 = 0 \iff r_{2,3} = -r_{3,3}(v_3 \cdot u_2)$

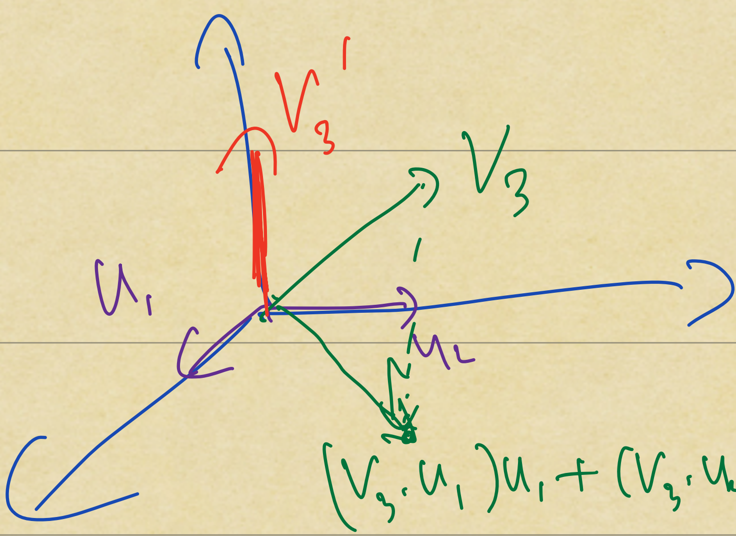
③ $u_3 \cdot u_3 = 1$

Initial guess: $r_{3,3} = 1 \Rightarrow r_{1,3} = -(v_3 \cdot u_1)$

$r_{2,3} = -(v_3 \cdot u_2)$

$$v_3' = -(v_3 \cdot u_1)u_1 - (v_3 \cdot u_2)u_2 + v_3$$

$$= v_3 - \underbrace{(v_3 \cdot u_1)u_1} - \underbrace{(v_3 \cdot u_2)u_2}$$




$(v_3 \cdot u_1)u_1 + (v_3 \cdot u_2)u_2$

$= s_3$

$$u_3 = \frac{v_3'}{\|v_3'\|}$$

So at each stage, having already found an orthonormal basis

$(u_1, \ldots, u_{m-1})$ for the span of

$(v_1, \ldots, v_{m-1})$ $= W_{m-1}$

We look at

$$s_m = \text{proj}_{W_{m-1}}(v_m)$$

$$= (v_m \cdot u_1)u_1 + (v_m \cdot u_2)u_2 + \ldots + (v_m \cdot u_{m-1})u_{m-1}$$

take $v_m' = v_m - s_m$

and then take $u_m = \dfrac{v_m'}{\|v_m'\|}$

## Example:

$$(v_1, v_2, v_3) = \left( \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right)$$

1) $$u_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$\Leftrightarrow \; v_1 = \sqrt{2}\, u_1$

2) $$s_2 = (v_2 \cdot u_1)\, u_1$$

$$= \left[ 1/\sqrt{2} + 1/\sqrt{2} \right] u_1$$

$$= \sqrt{2} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$-s_2$

$u_2 = \overbrace{-\sqrt{2}u_1 + v_2}$

$\Updownarrow$

$v_2 = \sqrt{2}u_1 + u_2$

(3) $v_2' = v_2 - s_2$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} := u_2$$

(4) $s_3 = (v_3 \cdot u_1)u_1 + (v_3 \cdot u_2)u_2$

$$= \frac{1}{\sqrt{2}} \cdot \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} + 0 \cdot u_2$$

$$= \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix}$$

(5) $v_3' = v_3 - s_3$

$$= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix}$$

$v_3' = v_3 - \frac{1}{\sqrt{2}} u_1$

$\Rightarrow v_3 = \frac{1}{\sqrt{2}} u_1 + v_3'$

$$= \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix}$$

(6) $$u_3 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$u_3 = \sqrt{2} \cdot v_3'$

$\Downarrow$

$v_3 = \frac{1}{\sqrt{2}} u_1 + \frac{1}{\sqrt{2}} u_3$

$$[u_1 \ u_2 \ u_3] = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \end{bmatrix}$$

$$\underbrace{[v_1 \ v_2 \ v_3]}_{A} = \underbrace{[u_1 \ u_2 \ u_3]}_{\substack{Q \\ \text{orthogonal}}} \underbrace{\begin{bmatrix} \sqrt{2} & \sqrt{2} & 1/\sqrt{2} \\ 0 & 1 & 0 \\ 0 & 0 & 1/\sqrt{2} \end{bmatrix}}_{\substack{R \\ \text{upper triangular}}}$$

This is the QR decomposition of the invertible matrix $A$

Rmk

$$Ax = v$$

$$\Updownarrow$$

$$QRx = v$$

$$\Updownarrow$$

$$Rx = Q^{-1}v$$

$$\Updownarrow$$

$$Rx = Q^{T}v$$